SONORA ARCHITECTURE & CLAUDE PROMPT SPEC

SONORA UI/UX Architecture & System Guide

A complete specification of the Aurora Nocturne design system, 4-stage psychological mood journey, fluid micro-interactions, and Claude engineering directives.

TARGET SYSTEM	ACTIVE BRANCH	DESIGN LANGUAGE	DATE
Claude 3.7 / 3.5 Sonnet	Ash (hack-3333)	Aurora Nocturne	November 2026

1. Product Vision & Executive Overview

SONORA is an AI-native emotive music discovery and mood transition engine. Traditional music platforms depend on generic genre buckets or static query matching. SONORA treats music search as an emotional dialogue, taking natural language descriptions, facial webcam telemetry, and 4-stage psychological transitions to curate hyper-relevant soundscapes via Google Gemini and Spotify.

🌌 Aurora Nocturne Design System

Dynamic 3-layer generative canvas with dark void glassmorphism, responsive ambient glow matching user mood intent, and crisp high-contrast white typography.

⚡ Fluid Micro-Interactions

Framer Motion spring physics for pill-based tab navigation (`layoutId="activeTabBadge"`) and seamless fade/slide transitions when switching between modules.

🌉 4-Stage Mood Transition

Constructs a psycholinguistic bridge from a starting emotion (e.g., *Melancholy*) to a target emotion (e.g., *Serene Joy*) across 4 structured stepping stones.

🟢 Native Spotify Integration

Direct track redirection to Spotify, bypassing fragile in-app web players and Spotify guest-mode playlist creation restrictions for a zero-friction listening experience.

2. "Aurora Nocturne" Design System Specifications

The interface uses a dark, luminous, translucent visual hierarchy built around 3 primary color tokens and layered glass effects:

Token Name	Hex / RGBA	Tailwind Class	Role & Psychological Semantics
Ink Void	`#07070D`	`bg-ink`	Primary infinite depth background; grounds neon glows
Deep Glass	`#12121E` (80% opacity)	`bg-surface/80`	Floating panels, search composer, cards with `backdrop-blur-xl`
Electric Violet	`#8B5CF6`	`text-violet`, `bg-violet`	Introspection, deep thinking, creative contemplation
Cyan Aurora	`#06B6D4`	`text-cyan`, `bg-cyan`	Clarity, calm focus, transition arrival, fresh energy
Neon Magenta	`#EC4899`	`text-magenta`, `bg-magenta`	Intense passion, high energy, vibrancy, euphoric moods
Spotify Emerald	`#10B981` / `#1DB954`	`text-emerald-400`	Audio actions, external play redirects, verified connectivity

Typography Rules

- **Headings & Wordmarks:** `Sora` / `Space Grotesk` (`font-display font-extrabold`) with letter-spacing `tracking-tight`.
- **Composer Prompt & Inputs:** Upgraded to `text-white font-medium` with `placeholder:text-white/80`. *Crucial improvement: avoids dim or unreadable gray placeholders over dark ambient backdrops.*
- **Technical Metadata & Counters:** `JetBrains Mono` (`font-mono text-xs`) for durations, character limits (`28/300`), and stage indices (`01 — Starting Mood`).

3. Component Hierarchy & Key UI/UX Evolutions

A. TopBar Navigation (`TopBar.tsx`)

> **Streamlined, Focused Top Navigation**
>
> Removed distracting status pills (Live/Mock mode pill) and redundant Spotify authentication buttons from the top bar. Instead, the top navigation focuses strictly on: 1. Brand Wordmark with animated soundwave bars. 2. Fluid center navigation pills between
> **Discover**
> and
> **Mood Transition**
> . 3. Contextual
> **Taste Preferences**
> trigger (visible only when in Discover mode).

The tab pill uses Framer Motion's `layoutId="activeTabBadge"`:

```
// Sliding pill background indicator across tabs
{activeTab === 'discover' && (
  <motion.span
    layoutId="activeTabBadge"
    className="absolute inset-0 rounded-full bg-gradient-to-r from-violet/90 to-magenta/80 shadow-m
    transition={{ type: 'spring', bounce: 0.18, duration: 0.45 }}
  />
)}
```

B. Main View Transition (`App.tsx`)

When switching between the Discover interface and Mood Transition page, the content smoothly animates without layout jank or unmounting delays:

```
<motion.div
  key={activeTab === 'transition' ? 'transition-view' : 'discover-view'}
  initial={{ opacity: 0, y: 14 }}
  animate={{ opacity: 1, y: 0 }}
  transition={{ duration: 0.28, ease: [0.16, 1, 0.3, 1] }}
  className="w-full flex-1 flex flex-col"
>
  {activeTab === 'transition' ? <MoodTransitionPage /> : <Hero ... />}
</motion.div>
```

C. Mood Transition Module (`LayeredPlaylistView.tsx`)

> **Architectural Decision: Removal of In-App Player & Playlist Creation**

> Spotify's Web API imposes severe sandbox limitations on creating playlists programmatically for guest/non-collaborative users. In-app audio previews also fail whenever Spotify removes 30-second preview URLs (a common recent Spotify API deprecation).
>
> **UX Improvement Applied:**
>
> - **Removed floating bottom audio player bar:** eliminates buggy audio element state and broken playback loops.
> - **Removed "Play Entire Journey" button:** prevents false expectations of bulk playlist creation.
> - **Direct Spotify Play Actions:** Every track row includes a prominent, animated Play button that redirects straight to `track.spotifyUrl` in a new tab with official metadata.
> - **Retained "New Journey" Flow:** Users can effortlessly re-run or re-tune their emotional trajectory.

4. The 4-Stage Psychological Emotional Model

SONORA guides users across a graduated continuum rather than an abrupt sonic change:

Stage	Phase Name	Color Accent	Curatorial Purpose
Stage 01	Starting Mood	Violet (`#8B5CF6`)	Validates the user's current feeling; empathetic resonance and acoustic familiarity.
Stage 02	First Transition	Indigo (`#6366F1`)	Introduces subtle tempo or instrumentation shifts; opens up emotional breathing room.
Stage 03	Second Transition	Cyan (`#06B6D4`)	Builds energetic momentum or harmonic warmth towards the desired state.
Stage 04	Destination Mood	Magenta (`#EC4899`)	Full sonic arrival into target state (euphoria, deep focus, relief, peacefulness).

5. Master Claude Prompt Template

Copy and paste the prompt below into Claude whenever you want Claude to collaborate, extend, refactor, or build new features for SONORA while adhering strictly to its architecture and design philosophy:

📋 SYSTEM PROMPT FOR CLAUDE: SONORA FRONTEND & UX ARCHITECT

You are the Lead UI/UX Engineer and Frontend Architect for **SONORA**, an AI-powered emotional music discovery platform. You write production-grade TypeScript, React 19, Tailwind CSS, and Framer Motion code.

Design System Guidelines ("Aurora Nocturne"):

- **Palette:** Void Ink (`#07070D`), Glass Surface (`#12121E/80`), Electric Violet (`#8B5CF6`), Cyan (`#06B6D4`), Magenta (`#EC4899`), and Spotify Emerald (`#10B981`).
- **Legibility & Contrast:** Always keep text inputs and critical labels in high-contrast white (`text-white font-medium` `placeholder:text-white/80`). Avoid muted or muddy grays over dark aurora backgrounds.
- **Glassmorphism:** Combine `backdrop-blur-xl` with translucent white hairline borders (`border-white/[0.08]` to `border-white/15`) and soft drop shadows.
- **Typography:** Sora / Space Grotesk for expressive display titles, Inter for body copy, JetBrains Mono for counters, durations, and technical labels.

Key Architectural Decisions to Maintain:

- **TopBar:** Keep it clean and streamlined. Only display the Wordmark, center fluid navigation tabs (Discover vs. Mood Transition), and contextual Taste Preferences button. Do NOT re-add mock mode pills or nav Spotify connect buttons.
- **Tab Animations:** Use Framer Motion's `layoutId="activeTabBadge"` on tab buttons for fluid spring-based pill gliding, and smooth entrance/exit animations on active views.
- **Mood Transition Playback:** Do NOT implement fragile custom HTML5 audio elements or "Play Entire Journey" playlist creation buttons (Spotify API guest limits). All play actions must open the track's `spotifyUrl` directly in a new tab (`target="_blank" rel="noopener noreferrer"`).
- **Testing & Type Safety:** Always preserve 100% passing tests (Vitest + React Testing Library) and strict TypeScript validation (`tsc --noEmit`). Never break existing integration test assertions.

6. Verification & Health Checklist

Verification Area	Command	Result
TypeScript Typecheck	`npm run typecheck`	✅ 0 errors across backend & frontend
Monorepo Unit & Integration Tests	`npm test`	✅ All 22 test suites (105 tests) passed
Mood Transition UI Suite	`frontend/tests/moodTransitionUI.test.tsx`	✅ 3/3 passed (Tab switch, mood validation, 4-stage flow)
Backend API Health	`GET /api/health`	✅ `{"ok": true}` on port 4000

Verification Area	Command	Result
Vite Dev Server	`http://localhost:5173`	✅ Active and hot-reloading on port 5173